Concurrency, Media and Security

13. Develop an Android application demonstrating Threads and Handlers for background processing.

Project Name: ThreadHandlerDemo

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
       android:id="@+id/btnStart"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Start Background Task"/>
   <TextView
       android:id="@+id/txtResult"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:textSize="22sp"
       android:paddingTop="20dp"/></LinearLayout>
```

MainActivity.java

```
package com.example.threadhandlerdemo;

import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnStart;
    TextView txtResult;

    Handler handler = new Handler(Looper.getMainLooper());

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnStart = findViewById(R.id.btnStart);
        txtResult = findViewById(R.id.txtResult);
```

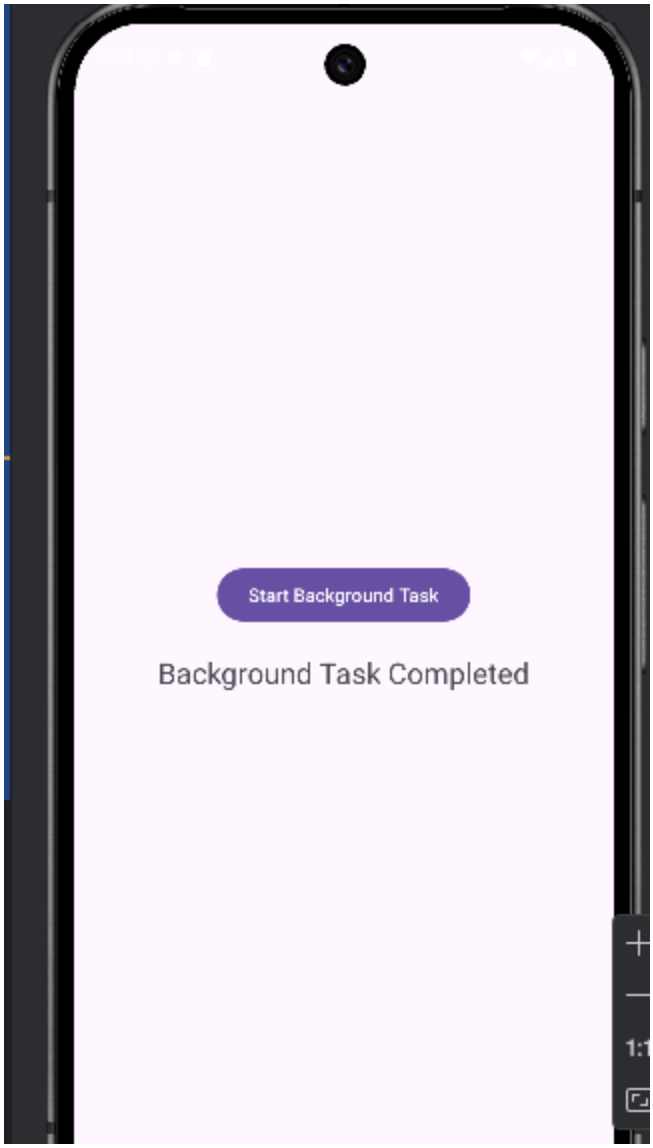
```
        btnStart.setOnClickListener(v -> {


            // Background Thread
            Thread thread = new Thread(() -> {
                try {
                    // Simulate Background Work
                    Thread.sleep(3000);


                } catch (Exception e) {
                    e.printStackTrace();
                }


                // Update UI using Handler
                handler.post(() -> {
                    txtResult.setText("Background Task Completed");
                });
            });


            thread.start();
        });
    }
}
```


Start Background Task
Background Task Completed